

Renormalization Invariance and Standing Preservation in the Non-Predictivity of Zero-Point Vacuum Energy

A Structural Equivalence for the Cosmological Constant Problem

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Abstract

A common presentation of the cosmological constant problem compares a cutoff-regulated zero-point energy density with the observed cosmological constant and treats the mismatch as a failed prediction of quantum field theory. The analysis is restricted to scalar quantities constructed from the homogeneous metric-proportional component of the vacuum stress tensor on a fixed background class, where the relevant renormalization freedom reduces to additive metric-proportional shifts. A physics-native analysis blocks the inference by noting that the absolute zero-point coefficient is not invariant under finite renormalization of this homogeneous component of the stress tensor. A structural analysis blocks the same inference by requiring standing preservation under admissible redescription. This paper gives a compact bridge between the two formulations. In the restricted sector, renormalization invariance and standing preservation are shown to be equivalent constraint conditions. Under the correspondence between admissible finite renormalization/covariant transformations preserving empirical normalization conditions and structural standing transformations, a quantity has standing if and only if it descends to the renormalization quotient that defines standalone predictive content. The zero-point vacuum-energy coefficient fails both tests for the same reason: it is a global additive offset without an invariant relational witness. In semiclassical gravity, compensating shifts of the matter vacuum term and the cosmological counterterm preserve the total effective cosmological parameter, so the split is not independently identifiable. The result does not remove the naturalness or matching problem for the measured value of the effective cosmological constant. It identifies the common structural reason why the zero-point summation procedure is not a cosmological prediction-forming map. The equivalence is established only for scalar quantities in the homogeneous vacuum sector, where the relevant renormalization freedom reduces to additive metric-proportional shifts.

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1 Purpose

There are two complementary ways to formulate the non-predictivity of zero-point vacuum energy. The first is native to renormalized quantum field theory (QFT): the homogeneous metric-proportional component of the vacuum stress tensor can be shifted by a finite local counterterm, and the split between matter vacuum energy and the cosmological counterterm is not separately identifiable in semiclassical gravity. The second is structural: an object has physical standing only when it remains stable under admissible redescrptions that preserve the witness ledger of the theory. A global additive offset with no invariant relational witness fails that condition.

This paper is a bridge between those formulations. It does not re-prove the full QFT argument, and it does not re-develop the full AASC closure machinery. Its narrow aim is to prove an equivalence statement:

In the homogeneous vacuum sector relevant to the usual zero-point comparison, the QFT requirement of renormalization-invariant standalone predictive content and the AASC requirement of standing preservation are the same quotient condition under the correspondence defined here.

The equivalence matters because it prevents two misreadings. From the QFT side, it shows that the failure of the zero-point estimate is not a discretionary convention but the local expression of a general invariance constraint. From the structural side, it shows that the AASC language is not imposing an external rule on physics; in this sector it binds to the same quotient structure already enforced by renormalization and semiclassical non-identifiability.

The result is deliberately scoped. It concerns the absolute zero-point coefficient of a single fixed homogeneous vacuum sector. It does not classify the full stress tensor in arbitrary curved spacetimes, does not deny curvature-dependent quantum effects, does not deny Casimir forces or phase-transition energy differences, and does not solve the naturalness problem associated with the small measured value of the effective cosmological constant. The target is only the direct comparison between a regulator-dependent zero-point sum and the observed cosmological constant as if the former were a standalone QFT prediction. Standard discussions of the cosmological constant problem, curved-spacetime stress-tensor renormalization, effective field theory, and non-predictivity of vacuum energy provide the background context [1, 2, 3, 4, 5, 6, 7, 8, 11, 12].

2 The QFT quotient condition

For a relativistic field, the formal zero-point expression is usually written schematically as

$$\rho_{\text{zp}}(\Lambda_{\text{UV}}) = \frac{1}{2} \int^{\Lambda_{\text{UV}}} \frac{d^3k}{(2\pi)^3} \sqrt{k^2 + m^2} \sim \frac{\Lambda_{\text{UV}}^4}{16\pi^2} + \dots \quad (1)$$

The controversial step is not the calculation of a regulated term. It is the assignment of direct cosmological predictive status to its absolute value.

The stress tensor is a composite operator. In curved-spacetime QFT its renormalized definition is constrained by locality, covariance, and conservation, but it remains ambiguous up to finite local conserved counterterms [5, 6, 7, 8, 9, 10]. The only component needed here is the homogeneous metric-proportional vacuum sector

$$\langle T_{\mu\nu} \rangle_{0,R} = -\rho_R g_{\mu\nu}, \quad (2)$$

with finite redefinition

$$\rho_R \longmapsto \rho_R + \delta\rho. \quad (3)$$

This is the restricted sector denoted by $\mathcal{V}_0$. Curvature-dependent ambiguities, anomalies, boundary counterterms, material response, and state-dependent stress-tensor effects lie outside the classification performed here.

Definition 1 (Renormalization quotient in the homogeneous vacuum sector). Let $\mathcal{D}_{\text{QFT}}$ be the set of descriptions of the homogeneous vacuum sector, including the finite convention used to assign ρ_R . Let $\mathfrak{R}$ be the admissible finite shift group generated by Eq. (3), together with transformations that preserve the already fixed empirical normalization conditions for non-vacuum matter observables. Two descriptions are equivalent, written $D \sim_{\mathfrak{R}} D'$, when they differ only by such transformations.

Definition 2 (Standalone predictive content). A scalar assignment $X : \mathcal{D}_{\text{QFT}} \rightarrow \mathbb{R}$ is standalone predictive content of the restricted QFT sector only if it descends to the quotient by $\sim_{\mathfrak{R}}$:

$$X(D) = X(D') \quad \text{whenever} \quad D \sim_{\mathfrak{R}} D'. \quad (4)$$

A non-invariant scalar may still be a scheme coordinate or an empirically fixed renormalized parameter. It is not predicted as a standalone number by the sector itself.

The content of this definition is not that every physically used parameter must be an invariant number by itself. Renormalized parameters transform with conventions; predictions are invariant relations after normalization conditions are fixed. The point is narrower: a cutoff-dependent or convention-dependent zero-point coefficient cannot be compared with observation as though it had already descended to the predictive quotient.

3 The structural standing condition

The minimal structural side uses only four AASC notions.

Definition 3 (Structural descriptions and admissible transformations). Let $\mathcal{D}_S$ be a set of descriptions of a target system, and let $\mathfrak{A}$ be the class of admissible transformations: redescrptions that preserve the admitted witness ledger of the system. In physics-bound applications these include changes of convention, gauge, coordinates, regularization, and finite renormalization that do not alter invariant measured responses.

Definition 4 (Witnesses and standing). A witness is an invariant measurement, response, transition, comparison, or prediction record admitted by the system. A scalar assignment $Y : \mathcal{D}_S \rightarrow \mathbb{R}$ has standing only if it is stable under admissible transformations:

$$Y(S) = Y(S') \quad \text{whenever} \quad S' = aS \quad \text{for some} \quad a \in \mathfrak{A}. \quad (5)$$

Equivalently, Y is constant on standing-equivalence classes.

Definition 5 (Skin, relational content, and boundary input). A scalar coordinate is *skin* if it changes under admissible transformations while preserving the witness ledger. A quantity is *relational standing content* if it survives only as a difference, transition, force, response, or boundary-conditioned comparison. A parameter is *boundary input* if its value is fixed by empirical matching to a larger envelope rather than predicted by the interior construction.

In the vacuum-energy case, the structural question is whether the absolute zero-point offset of one fixed interior vacuum can be assigned standing. The standing condition says no unless an invariant witness distinguishes that offset from an admissible redescription.

4 Binding correspondence

The bridge is established by fixing a correspondence between the QFT quotient and the structural standing quotient. The following table is not heuristic ornamentation; it defines the operational binding used in the theorem.

Structural term	Physics-bound correspondent in the restricted sector
Admissible (finite renormalization and covariant) transformation	Finite renormalization, finite counterterm shift, and coordinate/gauge/normalization transformation preserving admitted empirical normalization conditions and invariant records
Standing-equivalence class	Equivalence class of descriptions modulo $\sim_{\mathfrak{R}}$, i.e. the quotient on which standalone predictive content is defined
Witness	Invariant measurement, response, transition frequency, force, scattering or correlation prediction, curvature coefficient, or empirically fixed normalization record
Skin	Scheme-dependent coordinate such as the absolute homogeneous matter-sector vacuum coefficient $\rho_{\mathbf{R}}$
Relational standing content	Differences, responses, forces, transition quantities, or boundary/phase-conditioned comparisons in which common offsets cancel
Boundary input	Empirically fixed effective parameter such as the total Λ_{eff} , not derived by summing interior zero-point modes

Proposition 1 (Binding consistency). *For scalar quantities constructed from the homogeneous metric-proportional component of the vacuum stress tensor in $\mathcal{V}_0$, invariance under admissible transformations in the structural sense coincides with invariance under finite renormalization and associated convention-preserving transformations in the QFT sense.*

Proof. By the correspondence above, the admissible transformations of the structural description are exactly those transformations that preserve the QFT witness ledger: fixed empirical normalizations, invariant response quantities, and the total gravitational coefficient after matching. In the homogeneous vacuum sector the relevant finite transformation is $\rho_{\mathbf{R}} \mapsto \rho_{\mathbf{R}} + \delta\rho$, together with compensating changes in scheme coordinates. A scalar assignment is stable under the structural transformations if and only if it is unchanged under those finite renormalization transformations, i.e. if and only if it descends to the quotient $\mathcal{D}_{\text{QFT}}/\sim_{\mathfrak{R}}$. In particular, the transformation $T_{\mu\nu} \rightarrow T_{\mu\nu} - Cg_{\mu\nu}$ in the vacuum sector, equivalently $\rho_{\mathbf{R}} \rightarrow \rho_{\mathbf{R}} + C$ with sign convention fixed by Eq. (2), preserves invariant observables while shifting the vacuum-energy coordinate. Hence invariance under admissible transformations excludes absolute offsets from the standing/predictive class. $\square$

The proposition is intentionally restricted. It does not claim that every term in the full curved-spacetime stress tensor is exhausted by a constant metric-proportional ambiguity. It claims only that, for the sector used in the standard zero-point comparison, the two equivalence relations coincide.

5 Equivalence theorem

Theorem 1 (Renormalization invariance–standing equivalence). *Under the binding correspondence of Section 4, and for scalar quantities constructed from the homogeneous metric-proportional component of the vacuum stress tensor on a fixed background class, the following two implications hold in the restricted sector $\mathcal{V}_0$:*

- (i) **Standing $\Rightarrow$ QFT invariance.** *If a scalar has AASC standing, then it descends to the QFT renormalization quotient $\mathcal{D}_{\text{QFT}}/\sim_{\mathfrak{R}}$ and is eligible as standalone predictive content, unless fixed merely as an empirical boundary input.*
- (ii) **QFT invariance $\Rightarrow$ standing.** *If a scalar descends to the QFT renormalization quotient, then it defines a standing-equivalence class under the admissible transformations defined by the correspondence.*
- (iii) **Equivalence.** *Consequently, within $\mathcal{V}_0$, a scalar quantity has AASC standing if and only if it is standalone predictive content of QFT in the sense of descending to the renormalization quotient.*

The equivalence is local to this homogeneous vacuum sector and does not classify the full renormalized stress tensor in arbitrary curved spacetimes.

Proof. We prove the two directions explicitly.

(Standing $\Rightarrow$ QFT invariance). Suppose that X has AASC standing. By definition, it is invariant under all admissible transformations that preserve the witness ledger. By Proposition 1, these transformations coincide in $\mathcal{V}_0$ with the finite renormalization and convention-preserving transformations generated by Eq. (3). Therefore $X(D) = X(D')$ whenever $D \sim_{\mathfrak{R}} D'$. Hence X descends to the QFT quotient and qualifies as standalone predictive content unless its value is instead fixed merely by empirical matching.

(QFT invariance $\Rightarrow$ standing). Conversely, suppose that X descends to $\mathcal{D}_{\text{QFT}}/\sim_{\mathfrak{R}}$. Then X is invariant under every finite renormalization and covariant convention-preserving transformation in the restricted sector. Under the correspondence of Section 4, those transformations are precisely the admissible transformations preserving the structural witness ledger. In particular, invariance under $T_{\mu\nu} \rightarrow T_{\mu\nu} - Cg_{\mu\nu}$, equivalently $\rho_{\text{R}} \rightarrow \rho_{\text{R}} + C$ in the sign convention of Eq. (2), induces an equivalence relation on scalar assignments by identifying descriptions that differ only by global offsets. This equivalence relation is the standing-equivalence class under the correspondence. Thus X is stable across standing-equivalence classes and has AASC standing. The two criteria therefore select the same scalar assignments in $\mathcal{V}_0$. $\square$

The theorem is not mathematically difficult; its role is classificatory. It identifies the exact equivalence relation under which the zero-point construction would have to be invariant in order to become a prediction, and it shows that the structural and QFT formulations impose the same quotient condition. The QFT non-predictivity result is recovered as the physical realization of the standing-preservation constraint under the correspondence defined above.

Corollary 1 (Prediction-form obstruction). *Any procedure that assigns physical meaning to a regulator-dependent global offset without an invariant relational structure fails to define a valid standalone prediction in the restricted homogeneous vacuum sector.*

Proof. A regulator-dependent global offset changes under admissible finite redefinitions that preserve the witness ledger. By Theorem 1, it has neither standing nor QFT quotient-invariant predictive content unless it is converted into a relational quantity or fixed as boundary input. $\square$

6 Application to zero-point vacuum energy

Apply the theorem to the absolute zero-point coefficient appearing in Eq. (1). Under the finite homogeneous shift,

$$\rho_R \longmapsto \rho_R + \delta\rho, \quad (6)$$

its value changes while the admitted non-vacuum witnesses remain fixed. Hence

$$\rho_R \notin \mathcal{O}_{\text{pred}}(\mathcal{V}_0), \quad (7)$$

where $\mathcal{O}_{\text{pred}}(\mathcal{V}_0)$ denotes the quotient-invariant standalone predictive content of the restricted sector. In structural language, ρ_R is skin: a coordinate on a description, not a standing-bearing scalar.

This does not imply that every vacuum-energy-like quantity is meaningless. The equivalence theorem yields a three-role classification:

- (i) **Skin:** the absolute homogeneous matter-sector offset ρ_R , which changes under admissible redescription;
- (ii) **Relational standing content:** differences, responses, transition quantities, Casimir forces, and phase-conditioned comparisons, where common offsets cancel or are fixed by the relational setup;
- (iii) **Boundary input:** the empirically matched total effective cosmological parameter, not derived by the zero-point summation procedure.

This classification is exhaustive with respect to quantities invariant under the admissible transformation class defined above. There is no fourth role in which the absolute offset of a single fixed vacuum sector becomes a standalone cosmological prediction.

6.1 Semiclassical gravity and non-identifiability

In semiclassical gravity one writes

$$G_{\mu\nu} + \Lambda_R g_{\mu\nu} = 8\pi G_R \langle T_{\mu\nu} \rangle_R. \quad (8)$$

For a homogeneous vacuum term $\langle T_{\mu\nu} \rangle_{0,R} = -\rho_R g_{\mu\nu}$, a finite shift is compensated by

$$\rho_R \mapsto \rho_R + \delta\rho, \quad \Lambda_R \mapsto \Lambda_R - 8\pi G_R \delta\rho. \quad (9)$$

The invariant coefficient is the total

$$\Lambda_{\text{eff}} = \Lambda_R + 8\pi G_R \rho_R, \quad (10)$$

with the sign convention fixed by Eq. (8). Curvature observations determine the total after empirical matching, not the split. Thus gravity does not rescue the absolute zero-point coefficient as an independent observable; it displays its non-identifiability.

6.2 Casimir and phase differences

Casimir forces are not counterexamples. A measured Casimir force is a relational response of a specified material and boundary system. It is invariant under shifts of the global vacuum-energy zero, even though formal zero-point expressions may appear in calculational representations [13, 14, 15, 16]. In the present classification, Casimir quantities belong to relational standing content, not to the absolute-offset role.

The same distinction applies to energy differences between dynamically connected phases or metastable vacua. Phase differences can enter tunneling rates, domain-wall dynamics, and cosmological histories [17, 18]. They are not measurements of the absolute zero of a single isolated vacuum sector. They survive because their physical role is relational, dynamical, or boundary-conditioned.

7 What the bridge does and does not claim

The bridge theorem says that two filters coincide in the sector at issue:

$$\text{standing preservation} \iff \text{renormalization-quotient invariance.} \quad (11)$$

It does not assert that QFT is derived from AASC, nor that the structural vocabulary replaces the standard physics. The structural formulation makes explicit an invariance constraint already implicit in the practice of renormalized QFT. The equivalence established here is local to the homogeneous vacuum sector and does not classify the full renormalized stress tensor in arbitrary curved spacetimes.

Nor does the theorem remove the technical-naturalness problem. The measured smallness of Λ_{eff} remains a real problem for effective-field-theory matching, radiative stability, and microphysical explanation [1, 4, 12]. The theorem removes only one invalid prediction-forming procedure: the comparison between a cutoff-dependent zero-point sum and the observed cosmological constant as if the former were already an invariant prediction.

This distinction also clarifies why proposed ultraviolet completions, cancellation mechanisms, or selection rules do not repair the zero-point comparison as a prediction. They may address the value of the renormalized effective cosmological parameter. They do not by themselves convert a non-invariant global offset into a standalone predictive observable.

8 Conclusion

The non-predictivity of zero-point vacuum energy can be stated in two languages. In QFT, the absolute homogeneous vacuum coefficient fails to descend to the renormalization quotient. In AASC terms, the same coefficient lacks standing because it is not stable under admissible redesciptions preserving the witness ledger. The bridge theorem shows that, in the restricted homogeneous vacuum sector relevant to the standard cosmological-constant comparison, these are not merely analogous claims. They are the same constraint condition under a defined correspondence. The equivalence established here is local to the homogeneous vacuum sector and does not classify the full renormalized stress tensor in arbitrary curved spacetimes.

The zero-point summation procedure therefore fails at the prediction-forming stage. It does not produce a standalone cosmological prediction whose mismatch with observation could be called a failed numerical prediction. What remains is the empirical and theoretical problem of the small matched value of Λ_{eff} , not an observational refutation of a regulator-dependent absolute offset.

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